A new lower bound for the mixed covering code $K(1, 7, 1)$

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Abstract

Let $K(1, 7, 1)$ be the minimum size of a code in $\mathbb{F}_3 \times \mathbb{F}_2^7$ with covering radius 1. The tables of mixed binary/ternary covering codes give $42 \le K(1, 7, 1) \le 48$. We prove $K(1, 7, 1) \ge 45$. Following the codeword-distribution method of Östergård and Blass, the space is divided into cells by fixing the ternary coordinate and two binary coordinates; a counting inequality leaves 24, 109 and 489 possible distributions of 42, 43 and 44 codewords over the cells, up to symmetry. Each is shown infeasible by integer programming with two independent solvers, and the list of distributions is checked by a separate brute-force enumeration.

1 Introduction

A code $C \subseteq \mathbb{F}_3^t \times \mathbb{F}_2^b$ has *covering radius* 1 if every word is within Hamming distance 1 of some codeword, and $K(t, b, 1)$ is the minimum size of such a code [2]. Bounds are collected in the tables of Kéri [3], following [6, 1]. For $K(1, 7, 1)$ the tables give

$$42 \le K(1, 7, 1) \le 48.$$

The lower bound is due to Lang, Quistorff and Schneider [4]. The upper bound is the elementary construction $K(1, 7, 1) \le 3K(0, 7, 1) = 3 \cdot 16$, the direct product of $\mathbb{F}_3$ with the binary Hamming code of length 7. We are not aware of any later improvement.

Theorem 1. *$K(1, 7, 1) \ge 45$.*

A code with fewer than M words can be enlarged to exactly M words without losing the covering property, so it suffices to show that no covering code with exactly 42, 43 or 44 words exists.

2 Method

Write words as (a, v) with $a \in \mathbb{F}_3$ and $v = (v_1, \ldots, v_7) \in \mathbb{F}_2^7$. For $0 \le m \le 7$, the *level-m cell* (a, x), $x \in \mathbb{F}_2^m$, is the set of 2^{7-m} words with ternary coordinate a and $(v_1, \ldots, v_m) = x$. For a code C let $M_{a,x} = |C \cap (a, x)|$.

Lemma 2. *If C has covering radius 1, then for every level-m cell (a, x), with $f = 7 - m$,*

$$(1 + f)\, M_{a,x} + \sum_{a' \ne a} M_{a',x} + \sum_{i=1}^{m} M_{a,\, x+e_i} \;\ge\; 2^f.$$

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Proof. The balls of radius 1 around the codewords cover the 2^f words of the cell. A codeword in the cell covers itself and its f neighbours in the cell, obtained by changing one free binary coordinate. A codeword in a cell (a', x) with $a' \neq a$, or $(a, x + e_i)$, differs from the cell in exactly one fixed coordinate, so its ball meets the cell in exactly one word. A codeword in any other cell differs from every word of the cell in at least two fixed coordinates, so its ball misses the cell. □

This is the inequality of Östergård and Blass [5] adapted to one ternary coordinate. The search for a code with M words has two steps.

1. **Distributions.** At level 0, list the triples (M_0, M_1, M_2) with sum M that satisfy Lemma 2; at levels 1 and 2, split each cell in two and keep the refinements that satisfy the lemma at the new level. Permuting the ternary symbols and permuting or complementing the fixed binary coordinates are isometries; at level 2 they form a group of order $3! \cdot 2^2 \cdot 2! = 48$, and one representative of each orbit is kept.

2. **Exact cases.** For each level-2 representative, decide by integer programming whether there is a covering code with exactly $M_{a,x}$ codewords in each of the 12 cells: one binary variable per word, one covering constraint per word, one count constraint per cell.

3 Computation

M	level-2 distributions	up to symmetry	Gurobi 13.0	CPLEX 22.2	SCIP 10.0
42	596	24	24 infeasible	24 infeasible	24 infeasible
43	3 936	109	109 infeasible	109 infeasible	109 infeasible
44	18 108	489	489 infeasible	489 infeasible	not completed

Table 1: Cases for $K(1, 7, 1)$. All solvers were run on the same models, built by code separate from the enumeration. SCIP is an additional check; at $M = 44$ it was too slow and was stopped.

Table 1 summarises the computation, single-threaded on one core of an Intel Core Ultra 7 255HX. The two main solvers are Gurobi 13.0 (through gurobipy, size-limited licence) and IBM CPLEX 22.2 (Community Edition, which allows the 384 variables and 396 constraints of these models). At $M = 44$, Gurobi took at most 73 seconds per case and 18 minutes in total, and CPLEX at most 4 minutes per case and 21 minutes in total. SCIP 10.0 agrees on every case at $M = 42$ and 43; two of those needed 26 and 42 minutes.

The following checks were made on the steps before the solvers.

- *Lemma 2 from the definition.* For levels 0, 1 and 2, the number of words of a cell whose ball contains a given word u was computed from Hamming distances alone; it depends only on the cell of u, and equals the coefficients in the lemma.

- *Distributions.* A separate brute-force enumeration of the level-2 distributions, imposing the computed inequalities at levels 0, 1 and 2, gives 596 distributions for $M = 42$, 3 936 for $M = 43$ and 18 108 for $M = 44$. With the 48 symmetries written out independently, each is equivalent to one of the representatives, and each representative occurs.

- *The exact model.* The 48-word record code satisfies the model for its own cell distribution, and the same code with one word removed does not.

- *Known values.* The same pipeline reports no code with 15 words for $K(1,5,1)$ and none with 23 words for $K(1,6,1)$, and finds codes with 16 and 24 words, which an independent brute-force checker confirms; these are the known values $K(1,5,1) = 16$ and $K(1,6,1) = 24$.

Remark 3. All three solvers use floating-point arithmetic, so each verdict is evidence rather than an exact certificate; they come from different teams and agree on every case each completed.

Remark 4. In a companion note [7] we show by computer that any covering code of radius 1 in $\mathbb{F}_3 \times \mathbb{F}_2^7$ with at most 47 words has trivial automorphism group. Together with Theorem 1, a code improving the upper bound 48 would have 45, 46 or 47 words and no symmetry.

Code and data. All programs and logs are in the public repository

`https://github.com/Noah-Parsons/covering-codes`

- distributions: `src/exact_dist.py`;
- exact cases with Gurobi, CPLEX or SCIP: `src/leaves_t1.py`;
- checks of the distribution step: `src/verify_t1_leaves.py`;
- logs: `results/leaves_K171_*.log`.

References

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